

DOCUMENT SUMMARIZATION & VOICE GENERATION

Overview

Organizations and individuals work with large volumes of documents such as reports, research papers, business documents, meeting notes, policy documents, and technical material. Reading and understanding lengthy documents manually can be time-consuming, particularly when users need to quickly identify the key information.

A document summarization system can reduce this effort by extracting the most important information and presenting it in a concise format. However, users may also want to consume the summarized information through audio, particularly when reviewing content while travelling, multitasking, or working with accessibility requirements.

Users may ask questions such as:

- “Summarize this 50-page business report into the key findings and recommendations.”
- “Give me an executive summary of this document.”
- “Identify the major risks and decisions mentioned in this report.”
- “Convert the summary into an audio briefing that I can listen to.”
- “Summarize this document and generate a short voice-based explanation.”

Build an enhanced document intelligence system that accepts documents, extracts and processes their content, generates structured summaries, validates the summary against the source document, and converts the final summary into an audio output.

Tasks

Task	Task Overview	Weightage
Task 1 – Document Ingestion, Processing & Knowledge Base	Build the document processing pipeline required to ingest different document types and prepare their content for summarization.	20%
Task 2 – Intelligent Document Summarization & Information Extraction	Generate concise, structured and context-aware summaries from uploaded documents.	25%
Task 3 – Summary Validation, Multi-Agent Analysis & Voice Generation	Extend the summarization system with specialized analysis, validation and text-to-speech capabilities.	25%
Task 4 – Evaluation & User Application	Evaluate summary and audio quality and provide a simple application for end users.	20%
Task 5 – Architecture, Documentation & Demonstration	Document the complete solution and demonstrate the working document intelligence application.	10%
Total		100%

Task Details

Task 1 – Document Ingestion, Processing & Knowledge Base

- Build a robust document ingestion pipeline capable of accepting the supported document formats.
- The solution should extract the document content, preserve useful structural information such as pages or sections, clean the extracted text and prepare it for downstream summarization.
- For large documents, the participant should implement an appropriate chunking strategy. A searchable knowledge base/vector index may be created where required for retrieval and source-grounded analysis.

Task 2 – Intelligent Document Summarization & Information Extraction

- Implement the core document summarization capability.
- The system should be able to generate a concise and meaningful summary from the uploaded document while retaining important information.

Depending on the document type, the output may include:

- Executive summary
 - Key findings
 - Important facts
 - Risks
 - Decisions
 - Recommendations
 - Action items
- The user should be able to specify the required focus or summary format where applicable.
- The generated summary should be grounded in the source document, with appropriate references to the relevant pages or sections.

Task 3 – Summary Validation, Multi-Agent Analysis & Voice Generation

- Extend the basic summarization system into a multi-stage document intelligence workflow.

The implementation should include:

- **Document Analysis Agent** – identifies important information and relevant sections.
- **Summarization Agent** – generates the structured summary.
- **Summary Validation Agent** – checks whether the generated summary is supported by the source document.
- **Quality/Evaluation Agent** – evaluates completeness, relevance, consistency and overall summary quality.
- **Agent handoff** – allows the appropriate agent to process or validate different stages of the workflow.
- **A2A communication** – enables agents to exchange findings where applicable.
- **Text-to-Speech generation** – converts the final validated summary into an audio briefing.

- The workflow should ensure that the voice output is generated from the **validated final summary**.

Task 4 – Evaluation & User Application

- Evaluate the quality of the generated summaries using representative documents.

The evaluation should consider:

- Relevance
- Completeness
- Factual consistency
- Groundedness
- Summary quality
- Validation accuracy
- The application should provide a simple interface where users can upload a document and:
 1. View the generated summary.
 2. Review key findings/information.
 3. Review source references where available.
 4. Listen to the generated audio summary.
 5. Provide feedback on the usefulness of the output.

Task 5 – Architecture, Documentation & Demonstration

- Provide complete technical documentation and demonstrate the working solution.
- The architecture should clearly show:
Document → Extraction → Processing/Chunking → Analysis → Summarization → Validation → Evaluation → Text-to-Speech → User
- The documentation should explain the major technical decisions, agent workflow, evaluation approach and design trade-offs.

Deliverables

1. Architecture Diagram

- Include a high-level view of your system in JPEG or PDF.
- Show how data is processed from the provided documents through to the generated summary and voice output.

2. Design

- Articulate system design decisions taken and the trade-offs made.

3. Full Executable Code (Microservice)

- Focus on clarity and modularity.
- Provide a README explaining:
 - Project setup (how to install or run)
 - Sample usage (e.g., a document and the resulting summary/audio output)

4. Panel Presentation (10 minutes)

- Demo of your working solution outlining the design and approach (8 minutes).
- QCA (2 minutes).

Dataset Name

GovReport – Long Document Summarization Dataset

Primary Link

[GovReport Dataset – Hugging Face](#)

Alternative Links

[GovReport Dataset – GitHub / Dataset Resources](#)

[GovReport – Paper / Research Information](#)

Format

JSON / Text

Key Fields

document, summary, document ID and associated metadata where available.

Reason

The GovReport dataset contains long government reports paired with human-written summaries. It is suitable for evaluating long-document summarization, source-grounded analysis, summary validation and quality evaluation. The documents can also be used as input to the text-to-speech component, where the generated and validated summaries are converted into audio.